27. Numerical Integration

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Last time: $\mathrm{Vol}(\mathcal{D}) = \int_C \phi(x)\,dx \approx \frac{1}{n}\sum_{i=1}^{n} \phi(x)$

where $\phi(x) = \begin{cases} 1, & \text{if } x \in \mathcal{D} \\ 0, & \text{if } x \notin \mathcal{D} \end{cases}$

$= \mathbb{E}[\phi(X)]$ where $X \sim \mathrm{Unif}(C)$

Now, we can pick a different function in place of $\phi(x)$

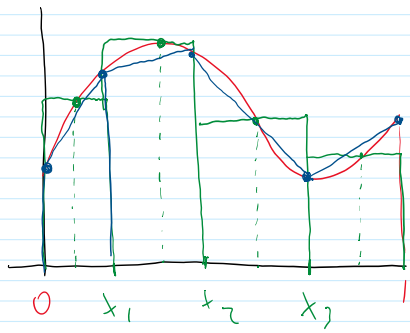
on $\mathbb{R}$, $\int_0^1 g(x)\,dx \approx \frac{1}{n}\sum_{i=1}^{n} g(x_i)$ for x_i "carefully" chosen pts in $[0,1]$

Deterministic Approaches $\Rightarrow \int_0^1 g(x)\,dx = \sum_{i=1}^{n} \underbrace{w_i}_{\text{weights}}\, \underbrace{g(x_i)}_{\text{points}}$

- Midpoint Rule $0 = x_0 < x_1 < \dots < x_n = 1$

$$\int_0^1 g(x)\,dx \approx \sum_{i=1}^{n} \underbrace{(x_i - x_{i-1})}_{\text{weight}}\; g\Big(\underbrace{\frac{x_i + x_{i-1}}{2}}_{\text{point}}\Big)$$

width $\times$ height



- Trapezoidal Rule

$$\int_0^1 g(x)\,dx \approx \sum_{i=1}^{n} (x_i - x_{i-1})\left(\frac{g(x_i) + g(x_{i-1})}{2}\right)$$

Monte Carlo Approaches

Law of Large #'s

$$\mu = \int_0^1 g(x)\,dx = \mathbb{E}[g(X)] \approx \frac{1}{n}\sum_{i=1}^{n} g(X_i) = \hat{\mu}$$

$X \sim \mathrm{Unif}[0,1]$ For $X_i \overset{iid}{\sim} \mathrm{Unif}[0,1]$

- our estimate $\hat{\mu}$ is now a random variable

$\rightarrow$ Mean: $\mathbb{E}\hat{\mu} = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} g(X_i)\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[g(X_i)] = \int_0^1 g(x)\,dx = \mu$

$\hat{\mu}$ is an unbiased est. for μ.

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$\rightarrow$ Variance: $Var\left(\frac{1}{n}\sum_{i=1}^{n} g(X_i)\right)$ $\qquad X \sim Unif[0,1]$

usual formula

$$= \frac{1}{n^2}\sum_{i=1}^{n} Var(g(X_i)) = \frac{1}{n} Var(g(X))$$

$$\sigma^2 = Var(g(X)) = \int_0^1 g(x)^2 dx - \mu^2$$

and $Var(\hat{\mu}) = \sigma^2/n$ $\leftarrow$ usual parametric rate

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Bayesian Interlude

Point is that parameters are now treated as Random Variables

- Before looking at data, we say that θ has a prior distribution $f(\theta)$
- After collecting data, say, X, we "update" our prior to get a posterior distribution for θ, $f(\theta \mid X)$

target to find

Estimator: $\hat{\theta} = E[\theta \mid X]$

* This Expectation may be very easy to solve if the prior distribution is chosen to make it easy to solve.

$\rightarrow$ However, it can be hard to solve!

I don't understand this example

Example: $X \sim N(\theta, 1)$, $\theta \sim Cauchy$

unknown mean $\qquad$ Known variance $\qquad$ prior

Posterior: $f(\theta \mid X) = \dfrac{f(X \mid \theta)\, f(\theta)}{f(X)}$

$$= \left[\cancel{\frac{1}{\sqrt{2\pi}}} \exp\left(-\frac{1}{2}(x-\theta)^2\right)\right]\left[\cancel{\frac{1}{\pi}} \frac{1}{1+\theta^2}\right]$$

$$= \frac{\left[\cancel{\sqrt{2\pi}}\ \exp\left(\ \ (x\ \ \ \theta)\ \right)\right]\left[\cancel{\pi}\ \ \ 1+\theta^2\right]}{\int_{-\infty}^{\infty} \cancel{\frac{1}{\sqrt{2\pi}}} \exp\left(-\frac{1}{2}(x-\theta)^2\right) \cancel{\frac{1}{\pi}} \frac{1}{1+\theta^2}\, d\theta}$$

Estimator: $\hat{\theta} = \mathbb{E}[\theta \mid X] = \int_{-\infty}^{\infty} \theta\, f(\theta \mid x)\, d\theta$

$$= \frac{\int_{-\infty}^{\infty} \left(\frac{\theta}{1+\theta^2}\right) \underbrace{e^{-\frac{1}{2}(x-\theta)^2}\, d\theta}_{\text{Integrate wrt Normal Distribution}}}{\int_{-\infty}^{\infty} \left(\frac{1}{1+\theta^2}\right) \underbrace{e^{-\frac{1}{2}(x-\theta)^2}\, d\theta}} \approx \frac{\cancel{\frac{1}{n}} \sum_{i=1}^{n} \frac{\theta_i}{1+\theta_i^2} \cancel{\sqrt{2\pi}}}{\cancel{\frac{1}{n}} \sum_{i=1}^{n} \frac{1}{1+\theta_i^2} \cancel{\sqrt{2\pi}}}$$

where $\theta_1, \dots, \theta_n \overset{iid}{\sim} N(x, 1)$

$$= \frac{\mathbb{E}\left[\frac{\theta}{1+\theta^2} \,\middle|\, x\right]}{\mathbb{E}\left[\frac{1}{1+\theta^2} \,\middle|\, x\right]}, \quad \theta \sim N(x, 1)$$